



A Machine Learning Approach for Sentiment Analysis to Nurture Mental Health Amidst COVID-19

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ABSTRACT

During the pandemic, when fresh news content is generated every minute about the widespread of the virus, many conversations revolve around the spread and cure of the contagion. At the hands of a commoner who posts news about COVID-19 on social media, the news may manifest itself to accommodate the said person's fear or negative propaganda which can potentially trigger a mass panic outbreak or can disrupt the mental health of a reader.

This research discusses the application of Machine Learning in Sentiment Analysis to classify Tweets about Coronavirus as fear sentiment or panic sentiment. It proposes the idea of a web-based application that caters to filter out the fear-inducing sentiment from a user's daily Twitter feed, thus giving the user accurate and well-spirited information.

Textual analysis is performed along with necessary textual data visualization. A substantial accuracy of 91% is achieved in the classification of brief Tweets using the Naïve Bayes method. An accuracy of 74% is achieved using the Logistic Regression classification method for brief tweets. This depicts the advancements in the field of sentimental analysis and sheds light on how it can be employed amidst a challenging situation like the pandemic to preserve mental health.

CCS CONCEPTS

• Computing methodologies—Machine learning • Machine learning approaches • Classification and regression trees

KEYWORDS

COVID-19, Coronavirus, pandemic, Naïve Bayes method, Textual analysis, Sentimental analysis, Python Machine Learning, R Statistical Software, Tweet-analysis, Machine Learning.

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1 Introduction

In the present era, every software application inclines towards Machine Learning to enhance its functionality, make it more user-friendly [1]. According to the Global Survey on Artificial Intelligence conducted by a management consulting firm, McKinsey [2], it is observed that a wide range of companies state that 20% of their earnings is accredited to Artificial Intelligence and Machine Learning. Machine learning comprises algorithms that enable computers to recognize patterns in tasks, learn their behavior, and henceforth automate them. Sentimental analysis is one such application of machine learning where algorithms are applied to detect the type of emotion behind text. Specifically, it is also used to analyze the emotion behind the text collected through a popular social media platform, Twitter, as noted by M. S. Neethu and R. Rajasree (2013) [3].

The year 2020 marks the beginning of a global health crisis, the Coronavirus outbreak. This has the people from all over the world within the haven of their homes, glued to a black screen to get a glimpse of what is going on outside their cozy abode. To communicate the happenings to their fellow beings, or to express their opinion about the pandemic, people resort to social media platforms. With the surge in the number of people being connected over social media, a single post, such as a tweet reaches out to billions of people. As noted in the survey conducted by Pew Research Center [4], in some countries, social media such as Facebook, Twitter are considered reliable news platforms! When

a single tweet has such an impact, it is important to ensure that this tweet does not instill fear or lead to a panic rise.

Unfortunately, social media platforms are being misused to trigger false fear amongst people about the virus. This panic would result in chaos and people being misled into buying hoax products that promise a cure [5]. An illustration of some panic triggering tweets is shown in Figure 1:

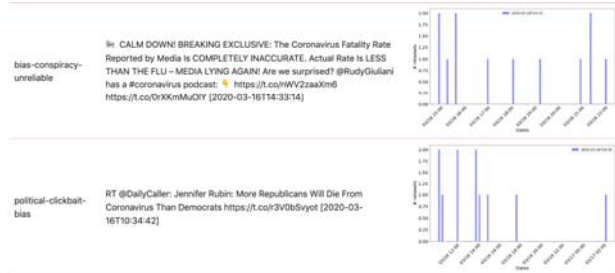


Figure 1: Sample fear-inducing tweets' growth over time [19]

The current understanding and knowledge of sentiment analysis of text data obtained through Twitter, a popular social media platform as illustrated by M. S. Neethu and R. Rajasree (2013) [3] is limited to performing classification of the Tweets according to emotions. Our current research converges this broad-spectrum into employing sentiment analysis to preserve mental health amidst the arising COVID-19 pandemic.

In this research, the implementation prospect of a web-based application is put forth. This application customizes a user's Twitter feed to view only the factual and encouraging tweets which would help the evasion or prevention of the virus rather than the ones that disrupt the mindset of the user.

2 Literature Survey

A study on the different approaches towards sentiment analysis revealed the application of numerous Machine Learning algorithms, all aimed at increasing the accuracy percentage of the classification results [6]. The impetus behind this study was to acquire the essence of various classification methods in Machine Learning which included- The Naïve Bayes algorithm and the Logistic Regression method. Besides the classification methods, there are additional methods employed for Sentiment analysis, the prominent of which are described below:

1. Support Vector Machines- This is a classification or regression algorithm whose paradigm is based on two key factors- Support Vectors and Hyperplane. A visual depiction aids the simplification of this concept (Figure 2). In the graph of Figure 2, the points on either side of the straight line are called Support Vectors and the latter is called Hyperplane. The support vectors essentially represent the items that are to be classified and the hyperplane is the line drawn to distinguish them. The crux of this algorithm is to determine the accurate location to place a hyperplane so that the classification is accurate. Balahur (2013) [7] used support

vector machines and supervised learning to classify tweets in an emotional context.

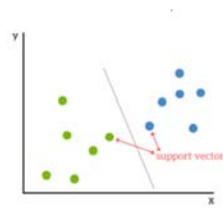


Figure 2: Theory behind Support Vector Machines [21]

2. The Recurrent Neural Network approach by Mikolov et al. (2010) [9][10] classifies the data with the help of an RNN model to achieve segregation in an emotional context.
3. There is an approach that uses Latent Dirichlet Allocation to analyze Twitter users' psychological reactions to COVID-19. [11]

A relatively simple yet, effective approach describes the use of descriptive textual analysis techniques and textual data visualization to detect the sentiment behind brief tweets by Jim Samuel, G.G.Md. Nawaz Ali, Md. Mokhesur Rahman, Ek Esawi, Yana Samuel [12]. The analytics was performed using R software, a prominent data science tool. A fear curve was laid out as a result of Sentiment analysis (Figure 3). This depicts the gradual increase in the development of "fear" emotion within the tweets. The bars indicate the daily increase in the count of the tweets with the "fear" emotion. We use this approach of classification in our research to develop a web-based application that facilitates the distillation of negative tweets from a user's feed.

3 Data Collection Phase

A dataset containing more than 900000 tweets during the period from February 2020 to March 2020 was obtained using the Twitter API, based on the case insensitive keyword "Corona" tweeted within the location of the United States.

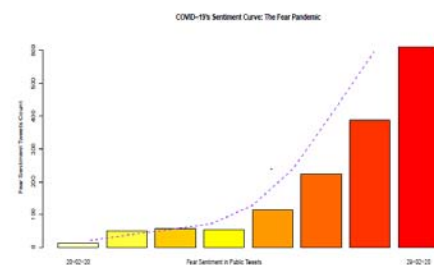


Figure 3: Increase in fear sentiment in tweets over time [12]

Now that data is collected, some common words (Unigrams), 2-word phrases (Bigram), 3-word phrases (Trigrams), 4-word phrases (Quadgrams) were extracted. The frequency of these phrases within the tweets was plotted for visualization (Figure 4).

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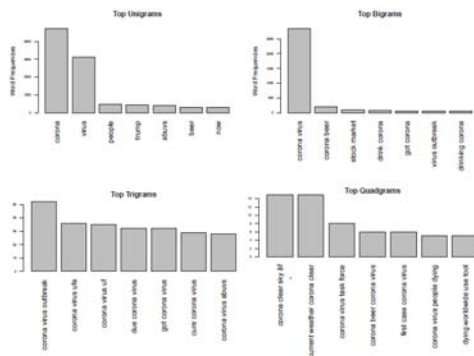


Figure 4: N-grams [12]

Tweet	Sentiment
This is wonderful! Mayo Clinic hometown Hilton hotel shows its support for health care workers during this trying time.	Positive
Great news: Save-on-Foods employees will receive a \$2-hour pay boost in response to the incredible work they are doing.	Positive
Let unite to fight against Covid 19. Listen to the government. Stay at home. Now is the best time to practice.	Positive
Erdo fan sends audio message to mobile users across Turkey in bid to raise awareness of #COVID19 among elderly.	Neutral
From Bergamo, Italy; of the worst hit hospitals by #COVID19 have a dear friend in Italy. She said it is hell.	Negative
With some of our students facing food insecurity, especially during this difficult time with #COVID19, please consider.	Negative

Figure 5: Examples of sentiments behind tweets [13]

It can be inferred from these graphs when the Coronavirus outbreak was in the early stages tweets were humor-oriented and words like “beer”, “now”, “trump” had higher frequency. The Trigrams and Quadgrams indicating fear such as “got corona virus”, “cure corona virus”, “corona virus people dying” and “dying worldwide use tool” see significant growth each day and this aligns to Figure 3 which indicated an increase in the fear emotion.

4 Implementation of the Algorithms

The entire procedure of Sentiment analysis is captured in Figure 6 [14]. Sentiment analysis is performed by associating each keyword or phrase with a sentiment score and then calculating the total sentiment score of the tweet (Figure 5 illustrates sample classification). These scores are predefined in the built-in packages of R, viz. “Syuzhet” and “Sentimentr”. The scores range from -1 to +1 and are grouped into sentiment classes such as fear, anger, and sadness [13].



Figure 6: Sentiment analysis procedure [14]

After defining a sentiment analysis scoring mechanism, a Machine Learning classification algorithm called the Naïve Bayes algorithm [15] is applied to the dataset to classify the data elements into the respective sentiment classes. The Naïve Bayes algorithm calculates the probability of each element of the dataset belonging to a class based on the attributes of the element and classifies that element as belonging to that class for which it has the highest probability. For example, if there is a round object and two classes say “Apples”, “Oranges”, if the probability of the attribute “Red” of the object is greater than the “Orange” attribute, the object belongs to the apple class. In our case, the tweets belonging to the positive sentiment class were marked with a value 0, and the tweets in the negative sentiment class were marked with a value 1. The dataset was divided into two halves, one with a character length less than 77 and the other half with the same value less than 120 to gauge the dependency of classification accuracy on the length of the dataset. Both halves contained 70 elements each. The results obtained are shown in Table 1:

Tweets (nchar < 77)			Tweets (nchar < 120)		
Negative		Positive	Negative		Positive
Negative	34	1	Negative	34	1
Positive	5	30	Positive	29	6
Accuracy: 0.9143			Accuracy: 0.5714		

Table 1: Results using Naïve Bayes classification [12]

Table 1 can be interpreted to infer that the accuracy of classification was 91.43%. This represents a significant increase in the reliability of the classification results obtained, that is simply put, indicates that a tweet classified by our algorithm as a tweet inducing negative emotion is actually indicating a negative emotion. Accuracy in Machine Learning is defined as the number of correctly predicted data points [16] However, the accuracy substantially decreased with the increase in the number of characters and was about 50% accurate for a tweet of <120 characters. With the usage of additional models and cleansing techniques, this accuracy can be improved. Parameters called “Sensitivity” and “Specificity” help in measuring the specific accuracy of a classification method. Sensitivity is defined as the ratio of correctly identified positives to the total number of positives, and Specificity is defined as the ratio of correctly identified negatives to the total number of negatives [17]. In this case, Sensitivity was 0.86 the considering the brief tweets. It was 0.17 when the long tweets were considered. Specificity was 0.97 when both brief and long tweets are considered. Thus, the Naïve Bayes method performs better in classifying the negative (fear) sentiment. This indicates that through the approach conveyed by our research, the results obtained are sufficiently reliable and the classification of Tweets is true to its nature.

Logistic regression [8] is another probabilistic classification model. This means that the model has a certain number of parameters that are assigned values based on the input dataset features and this model outputs binary classification. This is also a supervised model, meaning we train the dataset values previously

using a sigmoid function to fit a sigmoid curve. This curve is further used to classify the elements of the dataset into positive sentiment (1) and negative sentiment (0) about Coronavirus. These classes are similar to the Naïve Bayes method of classification. Like the previous method, the dataset was divided into two halves, based on the character length being <77 or <120 . The results obtained from this analysis were as recorded in Table 2.

Tweets (nchar < 77)			Tweets (nchar < 120)		
	Negative	Positive		Negative	Positive
Negative	30	5	Negative	21	14
Positive	13	22	Positive	19	16
Accuracy: 0.7429			Accuracy: 0.52		

Table 2. Results using Logistic Regression [12]

As evident from Table 2, the tweets with a lesser number of characters were more accurately classified (74.29% accuracy) than is the tweets with a greater number of characters (52% accuracy). Considering the previous approach and the current approach, the accuracy is lesser in comparison to the first approach proposed. However, the sensitivity was 0.63 considering brief tweets. It was 0.46 considering the long tweets. Specificity was 0.86 considering the brief Tweets. It was 0.60 considering the verbose Tweets. Thus, the logistic regression method performs well for a balanced classification [18].

Further analysis of the tweets dataset for the geographical location parameter, led to the following visualization of the fear sentiment data in Figure 7.

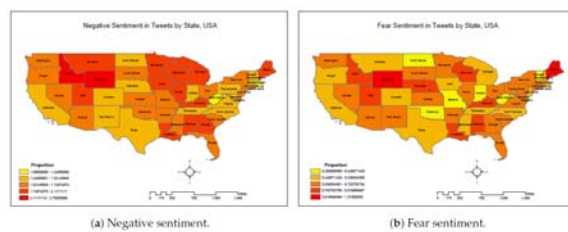


Figure 7: Mapping of the sentiment quotient to geography[12]

This data highlights the areas where fear sentiment was more prominent, and it is queer that these locations are different from the actual coronavirus hotspots where more people were affected. This can be accredited to the certitude that people in urban areas have better access to communication means to the tweets, thereby the fear sentiment prevails amongst these areas.

5 Proposed Methodology

Using the classification strategy described previously, an interface can be created for a user. This is a web application developed using the Django framework of Python. The choice of Django as

the framework for the UI is compelling because of the flexibility Django offers in terms of security, scalability, and seamless integration with the backend. Django is a web framework that follows the Model-View-Template architectural pattern. In this architectural pattern, the Model is the representation of the database tables in an object-oriented format using classes. The view consists of procedures or functions that render the objects of these models to the front-end templates. Templates are the scripts written for the browsers using HTML, CSS, and Bootstrap. The backend of the web application is the SQLite3 database. The development strategy of the web application is shown in Figure 8. The functionalities offered by this website would be as follows:

1. A user can create an account to personalize his/her Twitter account feed or use the application like a news source.
2. The interface is divided into two sections. The first section consists of Tweets that are promising, well-spirited, and ones that talk about the advancements in Vaccine development. This also consists of tweets about awareness and preventive measures.
3. The second section of the interface is the filtered Tweets which depict the emotions of fear, sadness, or anger. This will be displayed separately in this section and users can view this section only when they feel mentally comfortable doing so.
4. The data will be collected by using the Tweepy module of Python using suitable keywords by web scraping from Twitter API.
5. The Naïve Bayes algorithm is applied for classification as described in the implementation section.

This is the implementation perspective of the interface of the web application that uses Sentiment Analysis and Machine Learning to render an efficient mental health nurturing functionality.

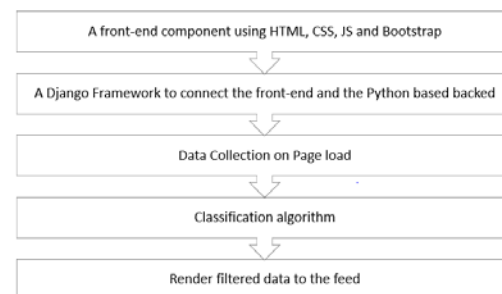


Figure 8: The proposed methodology

6 Conclusion

“Panic is highly contagious, especially in situations when nothing is known, and everything is in flux “- a quote by Stephen King portrays the negativity that is spread through such posts. As depicted, a thorough textual analysis of the data using the simplest, yet powerful Machine Learning models has fetched

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important detection results. The model can be extended to false information related to the virus and such posts can be filtered out, to make social media a more reliable platform for information sharing.



Figure 9: Government warning against fear-inducing news [5]

Social media is more about psychology and sociology than it is about technology. A flow of misinformation can not only disrupt the mental wellbeing of an individual but can wreak havoc in the balance of society if the panic leads to a mass outbreak (Figure 9, 10).

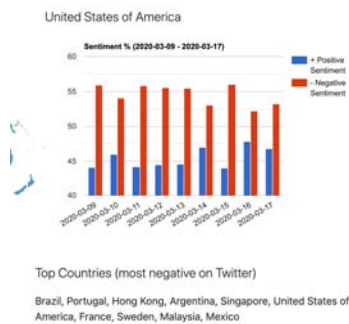


Figure 10: Negative sentiment spread [19]

The Naïve Bayes with logistic regression methods can also be used to produce the global pandemic insights report. With the onset of Big data analytics, voluminous unstructured data can be analyzed for patterns. It is significant to examine different social media applications with a focus on all textual aspects such as blogs, comments, images to derive a comprehensive aggregation. In the analysis of a disastrous situation, it is essential to plot sentiment against time, as depicted in the fear curve plot in Figure 3, and the geographical relevance, as illustrated in Figure 7.

Sentiment analysis is generally used by companies to gain insights about their products from the customers' point of view. This information helps them to align the products or marketing strategies in the direction of the consumer's liking. This article presented the present-day application of sentiment analysis amidst the pandemic crisis to scrutinize the perceptions of an individual as coping mechanisms and this information can be utilized to filter out the fear sentiment and instead provide the public with motivational and awareness-oriented information.

While the Government is focused on measures such as vaccines, treatment centers for the disease, providing food and shelter during Covid-19, there is another dimension of the pandemic that needs to be tended to. This is the mental health paradigm. Emotions such as fear, anxiety, stress are the common response mechanisms to global disaster. When an individual seeks out information on a social media platform, if posts indicating extreme panic and irrational decisions pop up, the anxiety level of the person is elevated. Social media companies are oriented towards monetization based on the popularity of the content and more often lack to determine the psychological impact of the content. Algorithms are designed to increase the engagement rate on social media but there is an exigency to filter content in the direction of emotional wellbeing.

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